

Carter Brown

Mechanical Engineer

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Education

University of California, Santa Barbara

Santa Barbara, CA

Bachelor of Science in Mechanical Engineering

2026

GPA: 3.75 / 4.00

Engineering Experience

Mechanical Engineer – Industry Capstone

2025–2026

Teledyne FLIR, University of California, Santa Barbara

Best Project Award

- Led thermal simulation for a GPS-denied drone navigation payload using COMSOL Multiphysics, reducing heat sink mass by 30% while keeping all components within safe operating temperatures under a 12W thermal load.
- Modeled three-dimensional conjugate heat transfer and airflow under operational flight conditions; simulation results closely matched experimental validation with a peak chip temperature of 77°C.
- Coordinated with electrical and software subteams to integrate mechanical housings, custom 4-layer PCBs, and software systems into a flight-ready payload.
- Managed project budget, handling all purchasing, vendor coordination, and expenditure tracking across five subteams.

Automatic Bike Chain Lubricator

Coursework

- Designed and built an automated chain lubrication system using a Hall Effect sensor to track mileage and a peristaltic pump to dispense oil every 150 miles; integrated Arduino logic, a digital display, and a custom 3D-printed housing.

Drone Frame Topology Optimization

Coursework

- Used COMSOL SIMP topology optimization to minimize a 3D-printable drone frame mass under multiple flight load cases (2g, 5g, 10g) while constraining maximum motor deflection to 2 mm.

Compressed Air Motor

Coursework

- Fabricated and assembled a functional compressed air motor using manual mill and lathe operations; applied engineering drawings and tolerancing to ensure reliable rotational performance.

Lab Experience

University of California, Santa Barbara

- Executed tensile, bending, and compression tests; generated stress-strain curves and determined material properties including Young's modulus, yield strength, and toughness.
- Conducted heat transfer and fluids experiments; analyzed steady and transient thermal data using MATLAB and evaluated pipe flow results using Bernoulli relations and Moody charts.
- Calibrated thermocouples, RTDs, strain gauges, and accelerometers; performed uncertainty analysis and signal conditioning.

Leadership and Work Experience

Valet

2022–Present

Park Hyatt Aviara, Carlsbad, CA

- Assisted guests in a high-volume luxury environment while safely handling over 100 vehicles per shift and maintaining efficient, incident-free operations.

Outside Services Lead

2020–2023

JC Golf, San Diego, CA

- Supervised daily operations for teams of up to ten staff members; trained new employees and coordinated schedules and task assignments.

Technical Skills

Software: COMSOL Multiphysics, SolidWorks, MATLAB, Arduino

Engineering: Thermal Analysis, CFD, Topology Optimization, Mechanical Design, System Integration

Manufacturing: Mill, Lathe, 3D Printing, Machine Shop Practices

Professional: Team Leadership, Budget Management